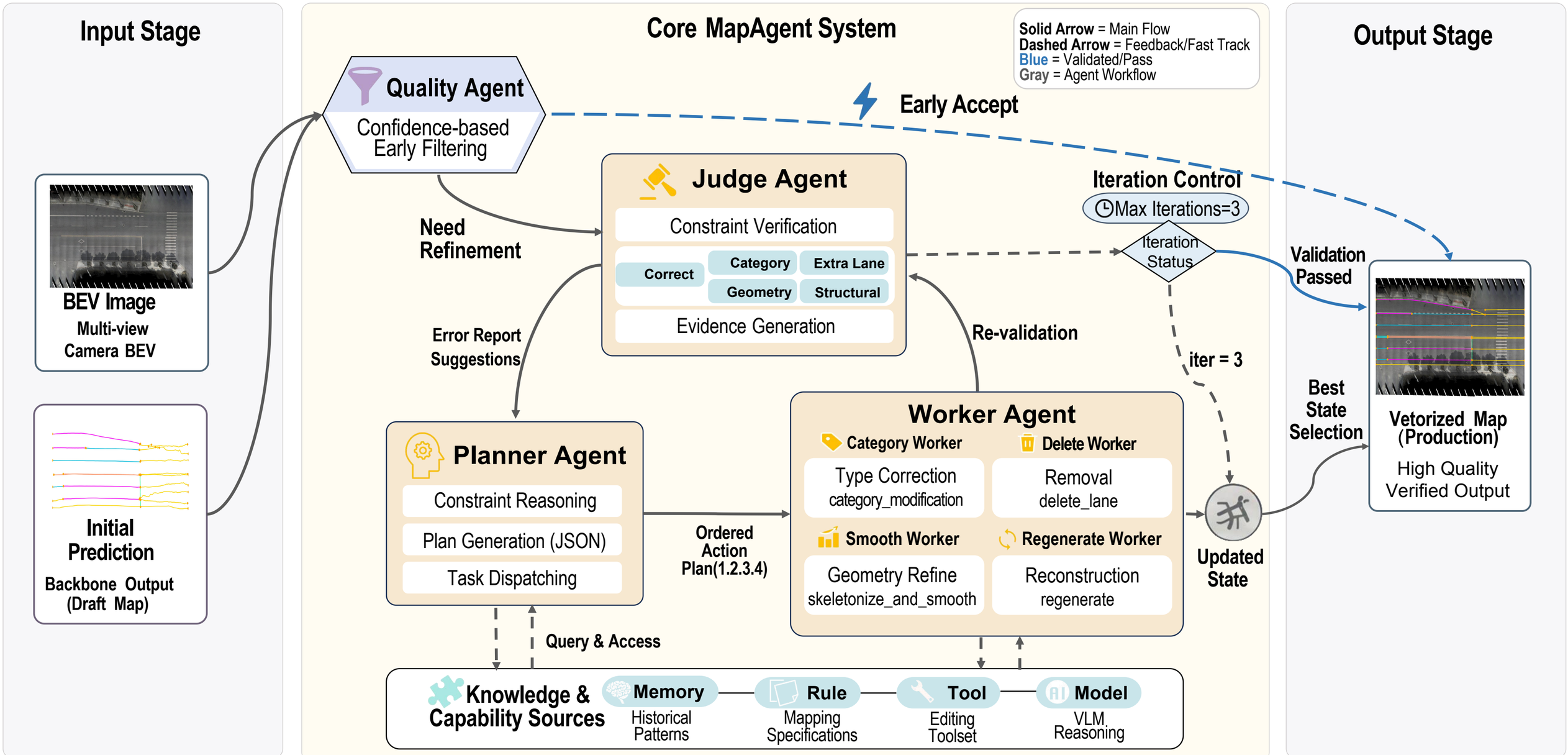
Input Stage
Core MapAgent System
Output Stage
Solid Arrow = Main Flow
Dashed Arrow = Feedback/Fast Track
Blue = Validated/Pass
Gray = Agent Workflow
Quality Agent
Confidence-based Early Filtering
Early Accept
BEV Image
Multi-view Camera BEV
Initial Prediction
Backbone Output (Draft Map)
Need Refinement
Judge Agent
Constraint Verification
Correct
Category
Extra Lane
Geometry
Structural
Evidence Generation
Iteration Control
Max Iterations=3
Iteration Status
Validation Passed
Error Report Suggestions
Re-validation
iter = 3
Best State Selection
Vetorized Map (Production)
High Quality Verified Output
Planner Agent
Constraint Reasoning
Plan Generation (JSON)
Task Dispatching
Ordered Action Plan(1.2.3.4)
Worker Agent
Category Worker
Type Correction
category_modification
Delete Worker
Removal
delete_lane
Smooth Worker
Geometry Refine
skeletonize_and_smooth
Regenerate Worker
Reconstruction
regenerate
Updated State
Query & Access
Knowledge & Capability Sources
Memory
Historical Patterns
Rule
Mapping Specifications
Tool
Editing Toolset
Model
VLM Reasoning